

LabBench Cheat Sheets

Compact reference tables for DSP, Control Systems,
Digital Logic, and Communication Systems.

Formulas, truth tables, and comparison charts — no prose, just the reference.

by **Dhananjay Kumar Seth**
labbench-hub.vercel.app

DSP Signal Lab — Cheat Sheet

Core formulas

Concept	Formula	Notes
Nyquist rate	$f_{\text{sample}} \geq 2 \times f_{\text{max}}$	Sample below this and you get aliasing
FFT bin resolution	$\Delta f = f_{\text{sample}} / N$	N = FFT size (e.g. 2048)
Filter cutoff	f_c	Where a filter's response is -3dB from passband
Q factor	$Q = f_c / \text{bandwidth}$	Higher Q = narrower, more resonant

Filter types at a glance

Type	Passes	Typical use
Lowpass (LPF)	below cutoff	Remove hiss/noise, smoothing
Highpass (HPF)	above cutoff	Remove DC offset / rumble
Bandpass (BPF)	band around center freq	Tune to one channel/station
Notch	everything except a narrow band	Remove 50/60Hz mains hum

Waveform harmonic content

Waveform	Harmonics present
Sine	fundamental only
Square	odd harmonics (f, 3f, 5f, 7f...), amplitude $\sim 1/n$
Sawtooth	odd + even harmonics, amplitude $\sim 1/n$
Triangle	odd harmonics only, amplitude $\sim 1/n^2$ (falls off faster than square)

PID Control Playground — Cheat Sheet

The control law

Term	Formula	Effect
Proportional	$K_p \cdot e(t)$	Reacts to current error; alone -> steady-state offset + oscillation
Integral	$K_i \cdot (\text{sum of } e(t))$	Kills steady-state error; too much -> slow oscillation (windup)
Derivative	$K_d \cdot (\text{rate of change of } e(t))$	Damps overshoot; too much -> noise-sensitive

Step-response metrics

Metric	Meaning
Rise time	How fast the output first reaches the target
Overshoot	How far past the target it swings before settling
Settling time	Time until output stays within a small band of target
Steady-state error	The gap that remains forever, if any

Tuning quick-reference

Symptom	Likely fix
Never reaches target, offset persists	Increase K_i
Oscillates and never settles	Decrease K_p , or increase K_d
Slow, sluggish response	Increase K_p
Fast but big overshoot	Increase K_d (adds damping)
Slow creeping oscillation	Decrease K_i (windup)

Logic Circuit Simulator & Waveform Viewer — Cheat Sheet

Gate truth tables

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

(NOT: output is the inverse of its single input — 0 -> 1, 1 -> 0)

Sequential building blocks

Element	Behavior
D Flip-Flop	Q captures D at every rising clock edge, holds until next edge
T Flip-Flop (clock divider)	Q toggles every rising edge -> divides clock frequency by 2
N-bit Counter	N flip-flops, ripple binary count $0..2^N-1$, one increment per edge
Shift Register	Serial input shifts through Q0 -> Q1 -> Q2... one stage per clock edge

Combinational vs. sequential

	Depends on	Has memory?	Examples
Combinational	current inputs only	No	AND/OR gates, adders, multiplexers
Sequential	current inputs + history	Yes	Flip-flops, counters, shift registers

Comms Simulator — Cheat Sheet

Analog modulation

Scheme	Formula	Key property
AM	$s(t) = [1 + k \cdot m(t)] \cdot \cos(2 \cdot \pi \cdot f_c \cdot t)$	$k > 1.0 \rightarrow$ over-modulation (envelope crosses zero)
FM	$s(t) = \cos(2 \cdot \pi \cdot f_c \cdot t + \beta \cdot \sin(2 \cdot \pi \cdot f_m \cdot t))$	Resistant to amplitude noise; β controls deviation

Digital modulation schemes

Scheme	Points	Bits/symbol	Noise robustness
BPSK	2	1	Highest (points maximally far apart)
QPSK	4	2	Medium
16-QAM	16	4	Lowest (points packed closer together)

BER & noise reference

Term	Meaning
AWGN	Additive White Gaussian Noise — the standard channel noise model
BER	Bit Error Rate — fraction of received bits that come out wrong
E_b/N_0 (SNR)	Energy-per-bit to noise-density ratio, usually in dB
$Q(x) = 0.5 \cdot \text{erfc}(x / \sqrt{2})$	The Gaussian Q-function — theoretical BER curve for BPSK/QPSK

labbench-hub.vercel.app

Thank you for supporting LabBench.